

CO2 AT-2

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SUB:CSA1501 CLOUD COMPUTING

**Q1. A media company wants to migrate its video streaming platform to the cloud. The system must handle millions of concurrent users with minimal buffering. Design a cloud architecture using scalability and caching techniques. Explain how load balancing and auto-scaling can ensure smooth streaming. Discuss cost optimization strategies for such high-demand workloads.**

### **Cloud Architecture Design**

The architecture should consist of:

- **Users** access the streaming platform through the Internet.
- A **Content Delivery Network (CDN)** caches videos close to users.
- A **Load Balancer** distributes requests across multiple application servers.
- **Auto-Scaling Groups** automatically add or remove servers depending on traffic.
- **Application Servers** process user authentication and streaming requests.
- **Cloud Storage** stores video files.
- **Database** stores user profiles, subscriptions, and viewing history.
- **Monitoring Services** continuously monitor system performance.

### **Scalability and Caching**

#### **Horizontal Scaling**

- Automatically launches additional virtual machines during high traffic.
- Removes unnecessary servers during low traffic.

#### **CDN Caching**

- Frequently accessed videos are cached at edge locations.
- Reduces latency and buffering.
- Decreases bandwidth usage.

#### **Application Caching**

- Store frequently accessed metadata in Redis or Memcached.

- Reduces database load.

## **Load Balancing**

Load balancing distributes incoming requests evenly among multiple servers.

Benefits:

- Prevents server overload.
- Ensures high availability.
- Reduces response time.
- Improves reliability.

## **Auto Scaling**

Auto Scaling monitors CPU utilization and network traffic.

Example:

- CPU >70% → Launch new instances.
- CPU <30% → Remove extra instances.

Benefits:

- Better performance.
- Automatic scaling.
- Lower operational cost.

## **Cost Optimization**

- Use **Auto Scaling** to avoid paying for unused servers.
- Store videos in **low-cost cloud object storage**.
- Use **CDN** to reduce bandwidth costs.
- Purchase **Reserved Instances** for predictable workloads.
- Use **Spot Instances** for background processing.
- Continuously monitor unused resources and remove them.

## **Conclusion**

The combination of CDN, load balancing, auto-scaling, and cloud storage enables the platform to support millions of users with minimal buffering while maintaining high availability and reducing operational costs.

**Q2. A fintech startup is deploying applications using containers across multiple cloud providers. They face challenges in maintaining consistency and managing deployments. Propose a solution using container orchestration and multi-cloud strategies. Explain how service discovery and scaling can be handled efficiently. Highlight the benefits and risks of a multi-cloud deployment model.**

### **Proposed Solution**

Deploy applications using **Docker containers** managed by **Kubernetes** across AWS, Azure, and Google Cloud.

Kubernetes provides:

- Automated deployment
- Scaling
- Load balancing
- Self-healing
- Centralized management

### **Multi-Cloud Strategy**

- Deploy Kubernetes clusters on each cloud provider.
- Use Infrastructure as Code (Terraform).
- Store container images in a shared container registry.
- Synchronize configurations using GitOps.

### **Service Discovery**

Use Kubernetes DNS and Services.

Each microservice receives a unique internal DNS name.

Benefits:

- Automatic discovery.
- Dynamic routing.

- Simplified communication.

## **Scaling**

Use:

- Horizontal Pod Autoscaler (HPA)
- Cluster Autoscaler

Scaling depends on:

- CPU utilization
- Memory usage
- Network traffic

## **Benefits of Multi-Cloud**

- High availability.
- Disaster recovery.
- Reduced vendor lock-in.
- Better global performance.
- Increased reliability.

## **Risks**

- Complex management.
- Different security policies.
- Higher networking costs.
- Configuration inconsistencies.
- Increased monitoring requirements.

## **Conclusion**

Kubernetes with multi-cloud deployment provides scalability, consistency, and fault tolerance while reducing vendor dependency. Proper orchestration and centralized monitoring minimize operational complexity.

**Q3. An e-learning platform experiences data loss due to accidental deletion in the cloud. The organization needs a robust backup and disaster recovery strategy. Design a solution using cloud-native backup, replication, and versioning. Explain how Recovery Time Objective (RTO) and Recovery Point Objective (RPO) are achieved. Provide steps to ensure business continuity during failures.**

### **Backup Strategy**

- Schedule automatic daily backups.
- Perform incremental backups every hour.
- Store backups in multiple geographic regions.
- Encrypt all backup data.

### **Replication**

Use cross-region replication.

Benefits:

- Data remains available if one region fails.
- Faster disaster recovery.

### **Versioning**

Enable object versioning.

Benefits:

- Recover accidentally deleted files.
- Restore previous versions.
- Protect against accidental overwrites.

### **Recovery Time Objective (RTO)**

**Definition:** Maximum acceptable downtime.

How achieved:

- Automatic failover.
- Standby servers.
- Cloud snapshots.

- Rapid restoration.

### **Recovery Point Objective (RPO)**

**Definition:** Maximum acceptable data loss.

How achieved:

- Continuous replication.
- Frequent backups.
- Database synchronization.

### **Business Continuity Steps**

1. Detect system failure.
2. Redirect traffic to backup region.
3. Restore latest backup.
4. Verify application functionality.
5. Resume normal operations.
6. Conduct post-incident analysis.

### **Conclusion**

Cloud-native backup, replication, and versioning minimize data loss, reduce downtime, and ensure continuous service during failures.

**Q4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data. Some applications must remain on-premises due to compliance requirements. Design a hybrid architecture that ensures secure communication between environments. Explain how identity management and encryption should be implemented. Discuss challenges in managing hybrid cloud environments.**

### **Hybrid Cloud Architecture**

#### **On-Premises**

Stores:

- Sensitive government databases.
- Confidential applications.

- Compliance-critical systems.

## **Public Cloud**

Hosts:

- Public websites.
- Citizen services.
- Analytics applications.
- Backup storage.

Both environments communicate using:

- Secure VPN
- Dedicated leased connection
- Private network gateway

## **Identity Management**

Implement a centralized Identity Provider.

Use:

- Single Sign-On (SSO)
- Multi-Factor Authentication (MFA)
- Role-Based Access Control (RBAC)

Benefits:

- Secure authentication.
- Consistent access control.
- Reduced administrative effort.

## **Encryption**

### **Data at Rest**

- Encrypt databases and storage using AES-256.

### **Data in Transit**

- Use TLS/SSL encryption.
- Secure VPN tunnels.

## **Key Management**

- Store encryption keys in a secure Key Management Service (KMS).

## **Challenges**

- Managing two infrastructures.
- Compliance requirements.
- Increased operational complexity.
- Network latency.
- Security policy synchronization.
- Higher maintenance costs.

## **Conclusion**

A hybrid cloud architecture combines the security of on-premises systems with the scalability of the public cloud while maintaining compliance through secure communication, centralized identity management, and strong encryption.

**Q5. A SaaS provider needs to ensure high availability and fault tolerance for its application. The system must continue operating even if one region goes down. Propose a multi-region deployment strategy in the cloud. Explain how data replication and failover mechanisms should be configured. Discuss monitoring and alerting strategies for proactive issue detection.**

### **Multi-Region Deployment Strategy**

Deploy identical application stacks in multiple cloud regions.

Each region contains:

- Load Balancer
- Application Servers
- Database
- Storage
- Monitoring Services

Use a Global DNS service to route users to the nearest healthy region.

## **Data Replication**

Implement:

- Synchronous replication for critical databases.
- Asynchronous replication for backup regions.
- Continuous storage replication.

Benefits:

- Prevents data loss.
- Improves disaster recovery.
- Ensures data consistency.

## **Failover Mechanism**

Configure automatic failover.

Steps:

1. Health checks monitor each region.
2. If a region fails, DNS redirects traffic.
3. Backup region immediately serves requests.
4. Failed region is restored and synchronized.

## **Monitoring and Alerting**

Monitor:

- CPU utilization
- Memory usage
- Disk utilization
- Network latency
- Application response time
- Error rates

Alert administrators using:

- Email
- SMS

- Mobile notifications
- Incident management tools

Use dashboards for real-time monitoring.

## **Conclusion**

A multi-region deployment with automated failover, continuous data replication, and proactive monitoring ensures high availability, fault tolerance, minimal downtime, and reliable service even during regional failures.